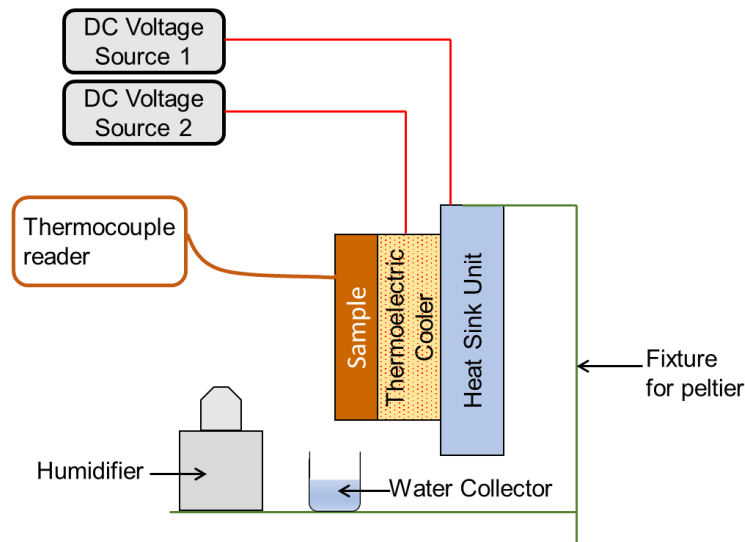


Condensation Dynamics on Metal Plates (Project Code: CD)

Objective: To investigate the rate of condensation on material and surface area (extended surfaces/finned surfaces)

Suggested setup schematic:



Schematic not to scale. This is a suggested setup, and you are free to make changes and improvements. Additional equipment/connections may be necessary.

The setup consists of a *thermoelectric Peltier cooler* and *copper/aluminum smooth/finned plates* as the test sample. The Peltier cooler is placed on a heat sink unit to facilitate lower surface temperatures. Both the Peltier cooler and the heat sink are connected to DC voltage sources. As the temperature of the Peltier cooler falls, the attached sample surface temperature drops below the dew point temperature, and condensation occurs. A collecting beaker at the bottom measures the rate of condensation (volume of water collected per unit time). The temperature of the sample surface is measured using thermocouples.

Requirement: The thermoelectric cooler (Peltier) should be 4 cm x 4 cm. The sample dimensions should be accordingly adjusted. The thickness of the aluminum and copper plates (without fins) can be ~ 2 mm.

Available budget: Rs. 25,000

Theory: The experimental setup, as shown above, will be designed and developed to study condensation dynamics. As the surface temperature falls below the dew point temperature, atmospheric vapor condenses on the surfaces. Such systems can be utilized for atmospheric water harvesting. The condensation rate can be controlled by controlling the Peltier temperature connected to the DC voltage source. Higher the temperature difference between the ambient and the Peltier, higher the rate of condensation. The rate of condensation and water collected can be quantified from the measured temperature of the surface. So, condensation heat flux for different condensation rates on different surfaces can be compared at the end.

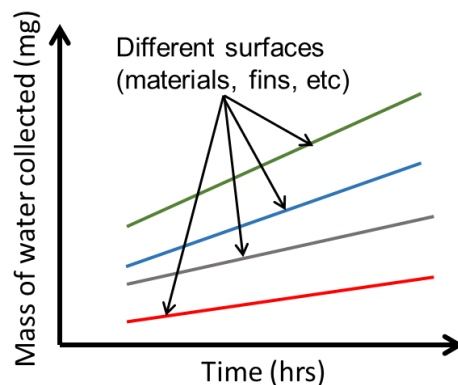
Suggested Finned Surfaces:

- 1) Copper Heat Sinks: <https://www.electronicshub.in/5-in-1-pure-copper-heat-sink-for-raspberry-4-model-b?search=heat%20sink>
- 2) Aluminum Heat Sinks: <https://www.electronicshub.in/aluminum-heat-sink-for-a4988-drv8825-stepper-motor-driver-5-pieces-pack?search=heat%20sink>
- 3) Aluminum and Copper Smooth Plates: (Mahalaxmi or our Machine Shop)

Suggestions for equipment (connections, fixtures, wires, thermal paste not included):

Items	Can be purchased from	Link/Contact
Glassware (for water collection)	Amazon	https://www.amazon.in/Spylx-Quality-Borosilicate-Glass-Beakers/dp/B08ZSK4QGX/
Aluminum Plate, Copper Plate	Mahalaxmi/ Amazon and our machine shop	Mahalaxmi contact: 9427416426 (Mr. Panchal),
Thermoelectric Cooler	ElectronicsComp	https://www.electronicshub.in/tec1-12706-thermoelectric-peltier-cooler-generator-module-india?search=TEC1-12706
Heat Sink Unit	Amazon	https://www.amazon.in/gp/product/B085G8WJCC/
Thermocouple/ thermocouple meter	Amazon	https://www.amazon.in/gp/product/B071V7T6TZ/
DC Voltage Sources	Amazon	
Humidifier	Amazon	https://www.amazon.in/Allin-Exporters-LP-2112M-Ultrasonic-Humidifier/dp/B0B3RM1NZP/
Power extension	Amazon/Flipkart	

Mandatory Experimental Results:



- 1) The mass of water collected versus time curve for (i) at least 4 different surfaces, and (ii) at least 2 different surface temperatures.
- 2) The condensation heat flux for each of the above cases.
- 3) The rate of condensation as a function of temperature.